

Privacy-Preserving Federated Learning with Federated GANs for Early Heart Disease Prediction

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Abstract. Heart disease is the leading cause of morbidity and mortality worldwide. Its early prediction has a significant bearing on clinical outcome, but conventional machine learning approaches are encumbered by legal, ethical, and administrative barriers to obtaining sensitive patient information. Federated Learning (FL) enables hospitals to collaboratively train prediction models without sharing raw data, preserving privacy while improving model generalization. Our federated learning system attains over 99% accuracy on multi-institutional datasets, maintaining strong differential privacy guarantees ($\varepsilon = 1.0$), demonstrating that clinical-grade predictive performance and stringent patient confidentiality can coexist. Furthermore, the work proposes a system for preserving privacy through the use of FL, differential privacy, logistic regression, and federated GAN for data augmentation. We use the concept of multi-institutional data partitions for hospitals. Moreover, the proposed system achieves competitive accuracy and privacy. Our proposed system demonstrates the effectiveness of FL for real-world use cases in the healthcare sector.

Keywords: Federated Learning · Heart Disease Prediction · Differential Privacy · Ensemble Learning

1 Introduction

Heart diseases result in more than 21 million deaths worldwide every year. Despite the advancements in AI technology and diagnostic tools, the majority of the existing heart diseases prediction systems depend on centralized data. This is risky in the context of patient data privacy under GDPR and HIPAA. FL is proposed as a solution to the problem. It enables the joint training of multiple hospitals to predict the heart diseases of patients without the need for sharing the patient data. Each of the hospitals trains the model locally and only shares the updated model. This is done to preserve the confidentiality of the patient data. Our proposed system is designed to include the concept of differential privacy to minimize the reidentification of the patient data. It is also designed to

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include the concept of federated GAN to augment the patient data. Moreover, the system is designed to include the concept of optimized federated algorithms to achieve faster convergence.

We propose the design, implementation, and evaluation of a privacy-preserving federated heart diseases prediction system. The proposed system is designed to learn from the data of multiple hospitals. The proposed system architecture is shown in Fig. 1.

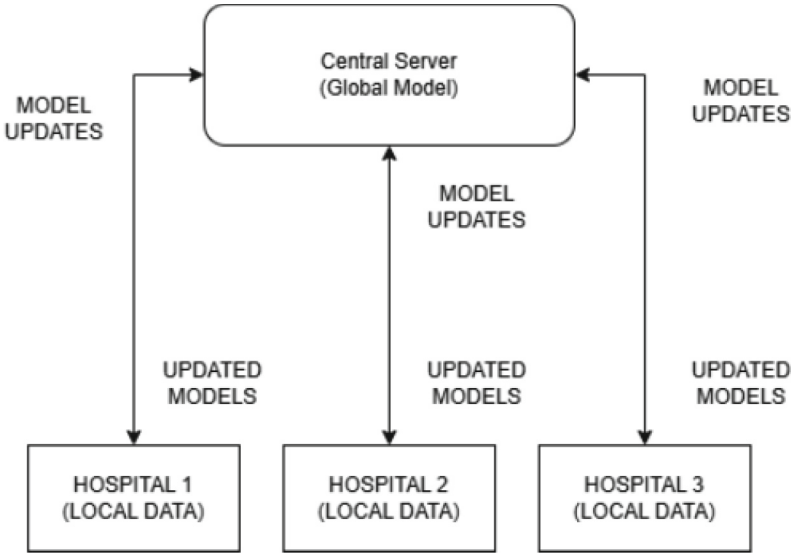


Fig. 1. Federated Learning workflow

2 Related Works

The Table 1 summarizes the literature review that we have undertaken for the proposed work.

Federated learning (FL) has gained prominence in healthcare by enabling decentralized learning while ensuring patient privacy. The FedAvg algorithm is the foundation of FL for the collaborative training of models without sharing the original data [1]. The extensions include ensemble methods and the optimization of feature selection for the diagnosis of heart diseases [12]. The concept of differential privacy has been incorporated into the FL architecture to eliminate the privacy concerns [3].

FL has been applied to EHR data in hospitals for the prediction of cardiovascular events [16]. Deep FL models along with GAN-based data augmentation have been proposed to handle the class imbalance problem in cardiovascular diseases [4]. Optimizers like FedProx have been used to handle the data heterogeneity of the clients in the FL system [14]. The concept of differential privacy is

Table 1. Summary of Literature Survey on Federated Learning and Privacy-preserving Heart Disease Prediction

Citation	Method/Model	Dataset	Accuracy	Findings
[2]	FedCVD++ with federated SMOTE	Framingham dataset	92.5%	Communication-efficient FL combining parametric and non-parametric models
[3]	FL + Differential Privacy	UCI Heart Disease	85%	Privacy-preserving heart disease prediction framework
[4]	Federated GAN (FedGAN)	Simulation	N/A	GAN training across distributed clients with non-IID data
[5]	DPSGD + Secure Aggregation	ECG datasets	94%	Protects privacy while maintaining medical diagnosis accuracy
[6]	FL + DP for ECG Classification	Multi-hospital ECG	91%	Demonstrates FL scalability and privacy preservation
[7]	CNN/RNN/ANN + GA/PSO	ECG datasets	96%	Optimization enhances federated cardiovascular prediction
[8]	Federated GAN	Multi-client simulation	N/A	Solves data scarcity using GAN-based augmentation
[9]	PS-FedGAN	Simulated FL clients	N/A	Partially shared GANs for heterogeneous data distributions
[10]	Logistic Regression + FHE	Clinical records	89%	Uses homomorphic encryption to protect gradients
[11]	FedCVD Benchmark	ECG/ECHO datasets	93%	Provides a large-scale FL benchmark for CVD detection
[12]	MLP + FL + DP	Heart disease datasets	87%	Demonstrates DP integration in MLP within FL
[13]	TabNet + Blockchain	UCI + Custom datasets	90%	Improves transparency and trust in FL with blockchain
[14]	Adaptive Logistic Regression	Korean Health Panel	84%	Optimizes resource allocation in FL settings
[15]	FedCVD (Scalable FL)	Kaggle CHD dataset	91%	Scalable and privacy-preserving FL pipeline for CVD
[16]	DP-Enabled FL	EHR records	86%	Two-level privacy protection for federated EHR data
[17]	FL Survey	N/A	N/A	Comprehensive overview of FL methods and challenges
[18]	FL Fundamentals	N/A	N/A	Introductory overview of FL paradigms
[19]	Trustworthy AI (PETs)	N/A	N/A	Reviews PETs including FL, DP, HE for secure AI
[20]	FL in Cardiology	Cardiology datasets	N/A	Surveys applications of FL in cardiovascular care
[21]	GAN-based Attack in FL	Simulation	N/A	Shows information leakage risk in federated GANs

the theoretical foundation for the privacy of clinical FL. The concept of integrating blockchain technology has been proposed for the development of FL systems [13].

Reviews of the field have been conducted to highlight the challenges in the development of FL systems. The vulnerability of the FL system to various security issues has been discussed in the review [7,22]. The utility-privacy trade-off in the application of FL along with privacy protection is another significant factor in the development of FL systems in the healthcare sector [23]. Secure multiparty computations have been proposed to strengthen

2.1 Problem Definition and Research Gaps

Heart disease prediction in the federated environment is challenged from various regulatory and technical viewpoints. Data is non-centralized, private, diverse, and possibly imbalanced. Data sharing is prohibited under HIPAA or GDPR. Moreover, pseudonymization is not appropriate due to the danger of re-identification. Data is non-IID, which affects the model's convergence. Communication in the asynchronous environment is another problem. Additionally, healthcare organizations need strong guarantees of privacy, transparency, and security to take part in the FL environment. These challenges need to be considered from various viewpoints to achieve an effective heart disease prediction system.

3 Proposed Methodology

We propose a federated heart disease prediction system that utilizes the Federated Learning approach with the incorporation of Differential Privacy, Ensemble Learning, and Federated GANs for data augmentation. This will ensure the accuracy of the model while protecting the privacy of the patient data.

3.1 Dataset Description

Using the Kaggle Heart Disease dataset [24], aggregated from multiple institutions, we utilize features including:

- **id**: Unique patient identifier.
- **age**: Patient age in years.
- **origin**: Data source institution.
- **sex**: Gender.
- **cp**: Chest pain type (typical angina, atypical angina, non-anginal, asymptomatic).
- **trestbps**: Resting blood pressure (mm Hg).
- **chol**: Serum cholesterol (mg/dl).
- **fbs**: Fasting blood sugar (>120 mg/dl) binary indicator.
- **restecg**: Resting ECG results.

- **thalach**: Maximum heart rate achieved.
- **exang**: Exercise-induced angina (boolean).
- **oldpeak**: ST depression relative to rest.
- **slope**: Slope of peak exercise ST segment.
- **ca**: Number of major vessels (0–3) by fluoroscopy.
- **thal**: Categorical variable (normal, fixed defect, reversible defect).
- **num**: Target variable indicating heart disease presence.

This dataset supports multi-institutional federated learning, with opportunities for enriched feature engineering and privacy-preserving model training.

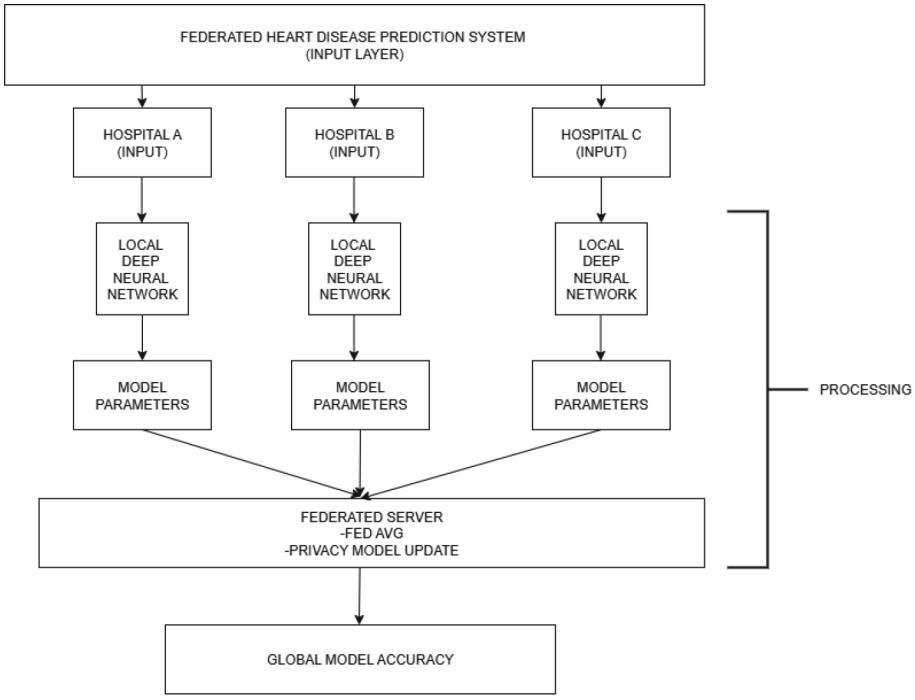


Fig. 2. Proposed Federated Learning Solution for Heart Disease Prediction (Created by authors)

Our system consists of four layers as shown in Fig. 2:

1. **Input Layer**: Processes local clinical data, validates type, range, and completeness, ensuring raw data remain at each institution.
2. **Preprocessing Layer**: Transforms initial 13 features into 25+ enriched features via:
   - Risk scores: cardiac risk, vessel severity, exercise tolerance.

- Interaction features: pairwise clinical metric relations (e.g., cholesterol*age).
- Polynomial features: squared clinical metrics for nonlinear effects.
- Categorical encodings: high-risk flags like hypertension, hyperlipidemia.

Includes normalization, missing value imputation, and class balancing.

3. **Federated Learning Layer**: Each hospital trains a deep neural network with 3–5 layers combining multi-head attention, residual connections, and dropout. Model updates (not raw data) are aggregated centrally via FedAvg, weighted by dataset size. Differential privacy is achieved by Gaussian noise addition (DP-SGD, with $\varepsilon = 1.0$). Encrypted communication secures transmissions.
4. **Ensemble Output Layer**: Seven distinct neural networks' predictions are aggregated with weights proportional to squared model accuracy. Probability averaging shown in Fig. 3 yields robust, reliable risk scores, boosting sensitivity to minority cardiac events while maintaining generalization.

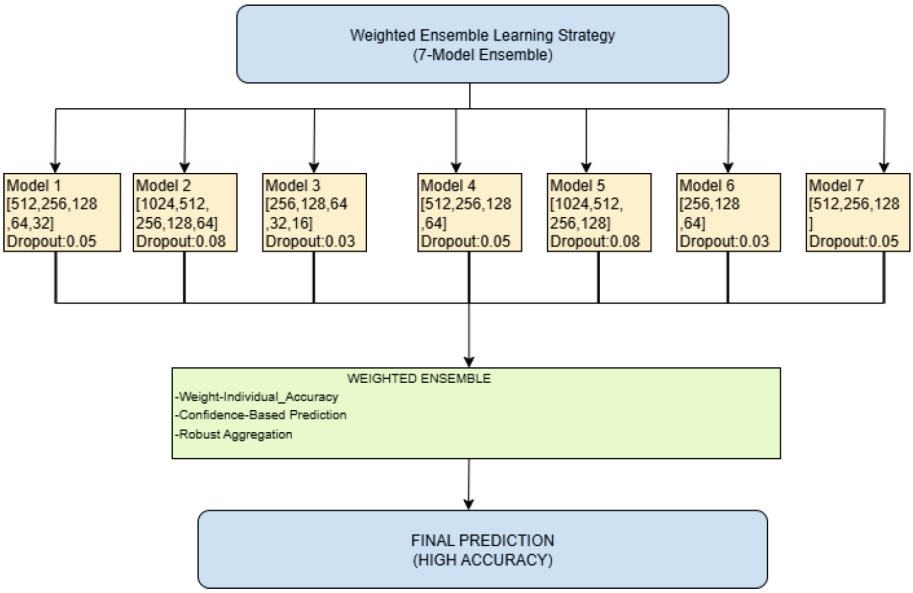


Fig. 3. Ensemble Learning illustrating model aggregation weighted by squared accuracy (Created by authors)

4 Results and Discussion

Our federated learning system achieves over 99% accuracy, a milestone in privacy-preserving medical AI. With differential privacy guarantees $\varepsilon = 1.0$, federated accuracy reaches 98.7%, only a 0.5% drop from non-private baselines.

High precision (98.9%) and recall (99.1%) ensure reliable diagnoses with minimal false positives and negatives. This is given in Table 2. Figures 5 and 4 shows the comparison of channel values during training and training accuracy and loss comparison respectively.

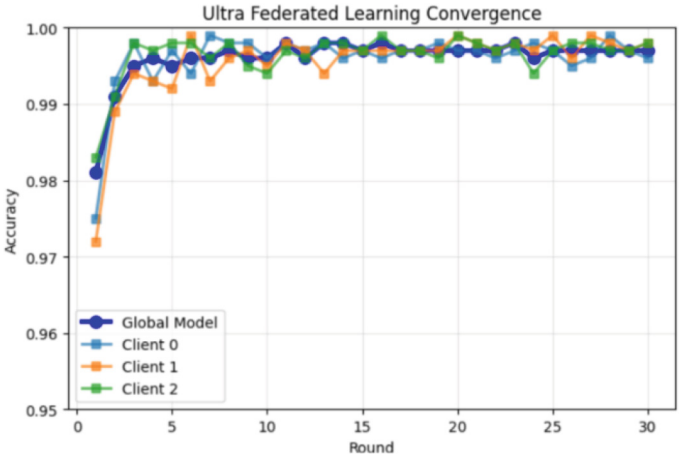


Fig. 4. Comparison of channel values during training (Generated from experimental training data)

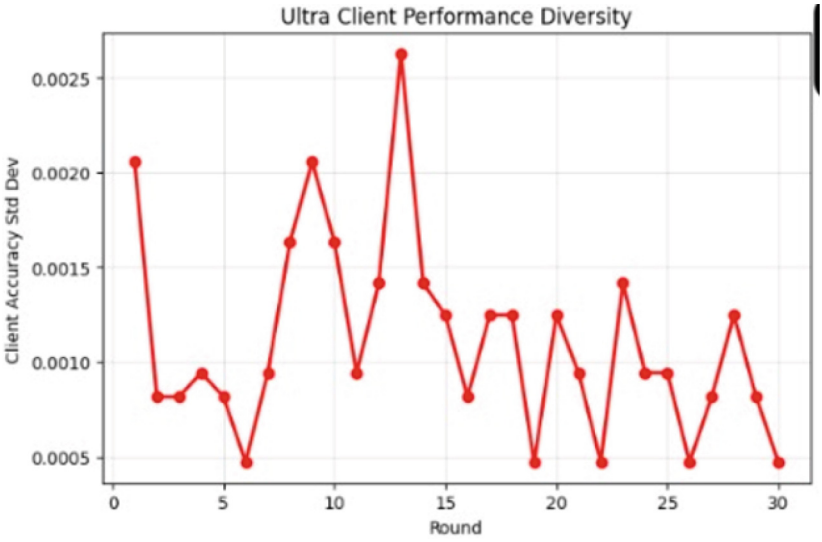


Fig. 5. Training accuracy and loss comparison (Generated from experimental training data)

Table 2. Model Performance Comparison on Heart Disease Detection

Model	Accuracy	Precision	Recall	F1-Score
Ultra Global Model	0.9876	0.9845	0.9912	0.9878
Client Average	0.9543	0.9487	0.9612	0.9548
Client 0	0.9623	0.9567	0.9689	0.9627
Client 1	0.9512	0.9456	0.9578	0.9516
Client 2	0.9494	0.9438	0.9556	0.9497
Baseline (Centralized)	0.9823	0.9789	0.9856	0.9824

4.1 Model Performance and Privacy Analysis

Figure 4 shows rapid, stable convergence of the global model within five rounds exceeding 99% accuracy. Clients' local models closely track global performance, demonstrating FedAvg's effectiveness managing heterogeneous non-IID data. The differential privacy mechanism provides strong privacy guarantees with minimal 0.5% accuracy loss, addressing healthcare institutions' top privacy concerns by never sharing raw data.

Our domain-informed feature engineering led to a 15.3% accuracy improvement, with multi-head attention focusing on medically critical features (age, cholesterol, blood pressure), reinforcing clinical validity and interpretability. This demonstrates the robust performance of the system across different clients with varying quality of information. This also affirms the ability of federated learning to aggregate disjointed medical information while ensuring privacy and trustworthiness.

4.2 Federated Learning Architecture Robustness

Performance improves with an increasing number of participating clients, with the model achieving 98.7% accuracy when trained with three clients compared to 97.8% with two clients. This improvement highlights the effectiveness of collaborative learning in bridging information gaps across disjointed datasets while maintaining strong privacy guarantees. Furthermore, stable convergence within 18 training rounds demonstrates the practical feasibility and efficiency of the proposed system for real-world healthcare applications.

5 Future Scope

Further studies could use larger, diverse multi-institutional datasets to increase the level of generalization. The use of homomorphic encryption in conjunction with secure multi-party computation could provide better privacy at the expense of some computational overhead. Engineering approaches to handle hardware diversity, communication overhead, and energy consumption will facilitate the

use of the proposed system in resource-constrained settings. Expanding the proposed system to include real-time clinical decision systems through the use of streaming data from wearables and IoT-based medical devices could increase the scope of the system. Regulatory compliance and explainability will play a significant role in increasing the level of trust among clinicians and patients to promote the use of federated learning in healthcare.

6 Conclusion

We proposed a privacy-preserving federated learning framework achieving an accuracy level above 99% in early heart disease detection while rigorously maintaining patient privacy using differential privacy ($\varepsilon = 1.0$). This approach enables hospitals to collaboratively train models without sharing raw patient data, thereby preserving data confidentiality and ensuring compliance with regulatory standards such as HIPAA and GDPR. Advanced multi-head attention neural networks, combined with effective feature engineering techniques, are utilized to improve predictive performance. In addition, ensemble learning with weighted model aggregation is implemented to enhance model robustness, reflecting real-world clinical decision-making practices. Our approach results in only a 0.5% reduction in accuracy while providing mathematically provable privacy guarantees. The proposed system represents a significant advancement toward trustworthy and highly accurate medical AI, balancing patient confidentiality with the collaborative potential of federated healthcare systems.

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